

Enterprise Digital Infrastructure – Web Lab2 Notes

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1. Caching: analyze the policies implemented by different servers with respect to caching and the behavior of the browser under different network conditions (i.e., throttling option in the Web Developer Tools)
2. Analysis of the HTTP/2 traffic: test different websites using the **nghttp** tool.

nghttp [OPTIONS]... <URI>...

```
Usage: nghttp [OPTIONS]... <URI>...
HTTP/2 client

<URI>      Specify URI to access.
Options:
-v, --verbose
            Print debug information such as reception and
            transmission of frames and name/value pairs. Specifying
            this option multiple times increases verbosity.
-n, --null-out
            Discard downloaded data.
-O, --remote-name
            Save download data in the current directory. The
            filename is derived from URI. If URI ends with '/',
            'index.html' is used as a filename. Not implemented
            yet.
-t, --timeout=<DURATION>
            Timeout each request after <DURATION>. Set 0 to disable
            timeout.
-w, --window-bits=<N>
            Sets the stream level initial window size to 2**<N>-1.
-W, --connection-window-bits=<N>
            Sets the connection level initial window size to
            2**<N>-1.
-a, --get-assets
            Download assets such as stylesheets, images and script
            files linked from the downloaded resource. Only links
            whose origins are the same with the linking resource
            will be downloaded. nghttp prioritizes resources using
            HTTP/2 dependency based priority. The priority order,
            from highest to lowest, is html itself, css, javascript
            and images.
-s, --stat  Print statistics.
-H, --header=<HEADER>
            Add a header to the requests. Example: -H':method: PUT'
--trailer=<HEADER>
            Add a trailer header to the requests. <HEADER> must not
            include pseudo header field (header field name starting
            with ':'). To send trailer, one must use -d option to
            send request body. Example: --trailer 'foo: bar'.
--cert=<CERT>
            Use the specified client certificate file. The file
            must be in PEM format.
--key=<KEY>
            Use the client private key file. The file must be in
            PEM format.
```

The man page provides many additional options that can be used with this command.

Example:

nghttp -sv https://poldo.unipv.it/

3. Analysis of the HTTP/3 traffic: visit a commercial website, check whether the web server supports HTTP/3 and analyze the traffic generated using Wireshark.
4. Benchmarking experiments: test the performance of different websites under HTTP/1.1 and HTTP/2 using benchmarking tools, such as **ab** and **h2load**.

```
Usage: ab [options] [http[s]://]hostname[:port]/path
Options are:
-n requests      Number of requests to perform
-c concurrency   Number of multiple requests to make at a time
-t timelimit     Seconds to max. to spend on benchmarking
                 This implies -n 50000
-s timeout       Seconds to max. wait for each response
                 Default is 30 seconds
-b windowsize    Size of TCP send/receive buffer, in bytes
-B address       Address to bind to when making outgoing connections
-p postfile      File containing data to POST. Remember also to set -T
-u putfile       File containing data to PUT. Remember also to set -T
-T content-type  Content-type header to use for POST/PUT data, eg.
                 'application/x-www-form-urlencoded'
                 Default is 'text/plain'
-v verbosity     How much troubleshooting info to print
-w              Print out results in HTML tables
-i              Use HEAD instead of GET
-x attributes    String to insert as table attributes
-y attributes    String to insert as tr attributes
-z attributes    String to insert as td or th attributes
-C attribute     Add cookie, eg. 'Apache=1234'. (repeatable)
-H attribute     Add Arbitrary header line, eg. 'Accept-Encoding: gzip'
                 Inserted after all normal header lines. (repeatable)
-A attribute     Add Basic WWW Authentication, the attributes
                 are a colon separated username and password.
-P attribute     Add Basic Proxy Authentication, the attributes
                 are a colon separated username and password.
-X proxy:port    Proxyserver and port number to use
-V              Print version number and exit
-k              Use HTTP KeepAlive feature
-d              Do not show percentiles served table.
-S              Do not show confidence estimators and warnings.
-q              Do not show progress when doing more than 150 requests
-l              Accept variable document length (use this for dynamic pages)
-g filename      Output collected data to gnuplot format file.
-e filename      Output CSV file with percentages served
-r              Don't exit on socket receive errors.
-m method        Method name
-h              Display usage information (this message)
-I              Disable TLS Server Name Indication (SNI) extension
-Z ciphersuite   Specify SSL/TLS cipher suite (See openssl ciphers)
-f protocol      Specify SSL/TLS protocol
                 (SSL2, TLS1, TLS1.1, TLS1.2, TLS1.3 or ALL)
-E certfile      Specify optional client certificate chain and private key
```

The man page provides detailed information about individual options.

Example:

ab https://poldo.unipv.it/

Usage: h2load [OPTIONS]... [URI]...
benchmarking tool for HTTP/2 server

<URI> Specify URI to access. Multiple URIs can be specified. URIs are used in this order for each client. All URIs are used, then first URI is used and then 2nd URI, and so on. The scheme, host and port in the subsequent URIs, if present, are ignored. Those in the first URI are used solely. Definition of a base URI overrides all scheme, host or port values.

Options:

-n, --requests=<N>
Number of requests across all clients. If it is used with --timing-script-file option, this option specifies the number of requests each client performs rather than the number of requests across all clients. This option is ignored if timing-based benchmarking is enabled (see --duration option).
Default: 1

-c, --clients=<N>
Number of concurrent clients. With -r option, this specifies the maximum number of connections to be made.
Default: 1

-t, --threads=<N>
Number of native threads.
Default: 1

-i, --input-file=<PATH>
Path of a file with multiple URIs are separated by EOLs. This option will disable URIs getting from command-line. If '-' is given as <PATH>, URIs will be read from stdin. URIs are used in this order for each client. All URIs are used, then first URI is used and then 2nd URI, and so on. The scheme, host and port in the subsequent URIs, if present, are ignored. Those in the first URI are used solely. Definition of a base URI overrides all scheme, host or port values.

-m, --max-concurrent-streams=<N>
Max concurrent streams to issue per session. When http/1.1 is used, this specifies the number of HTTP pipelining requests in-flight.
Default: 1

-f, --max-frame-size=<SIZE>
Maximum frame size that the local endpoint is willing to receive.
Default: 16K

The man page provides many addition options that can be used to customize the HTTP requests sent to the web server.

Example:

h2load https://poldo.unipv.it/